



ASSESSMENT:	GROUP ASSIGNMENT	PROGRAMME:	Diploma in ICT
MODULE CODE:	NDTA 631	MARKS:	100
MODULE DESCRIPTION:	Data Analysis And Visualization	DATE:	20 July 2026
EXAMINER(S):	Melvin Kisten	DUE DATE:	24 August 2026 (NO EXTENSIONS)
GROUP SIZE:	6 to 7 students		

INSTRUCTIONS:

- Answer ALL questions.
- Indicate your **full names and student number** on your **answer sheet, code, etc.**
- **Copying from a fellow student or any source will be treated as plagiarism.**
- **Submission Requirements**
 - **Report (7 to 9 pages) DOCX and PDF**
 - **Video demo (Optional but recommended)**
 - Create dedicated GitHub repository, include:
 - Comprehensive codebase or Jupyter Notebook or Python scripts
 - Execution instructions
 - Environment configuration
 - Maintain clean, commented code
 - Use version control (**git**)
 - Implement error handling

Group Assignment - Data Analysis and Visualisation:

Dataset(s)

Each group must pick at least **two** different **South African** datasets from World Bank Open Data:

<https://data360.worldbank.org/en/search?country=ZAF>

Your **two South African** datasets must work together to tell one clear story.

No two groups may use the same datasets.

Before starting, check what datasets other groups have chosen so you do not repeat the same datasets.

Else you will lose 15% of your final group mark!

Make a complete data analysis and visualisation pipeline:

- Load, clean, and prepare a min of 2 real datasets
- Use Python (Pandas, NumPy, Matplotlib, Seaborn) or Excel, etc
- Build and query a database
- Create clear visualisations
- Write an actionable report

Deliverables:

1. **Data Preparation (15 marks)**
 - Load data and clean it
 - Handle missing values
 - Generate descriptive stats and insights
2. **Numerical Analysis (20 marks)**
 - Use NumPy for calculations
 - Reshape arrays and perform operations
 - Explain findings
3. **Visualisation (20 marks)**
 - Create plots (bar, scatter, box, histograms, etc)
 - Explain trends and patterns
 - Use colour and labels clearly
4. **Database Integration (20 marks)**
 - Build and query a database

- Update and delete records safely
- Load database data into Pandas

5. Python/Excel Data Analysis (15 marks)

- Clean and transform data
- Apply conditional formatting
- Create charts and summarise findings

6. Report & Demo (10 marks)

- Write a 7-9 page report
- Include dataset details, methods, visualisations, and conclusions
- Present with clear demo

Marking Rubric (100 Marks)

Criterion	What to Look For	Score Range	Quick Check Questions
Data Preparation (15)	Do the students show how they cleaned the data, handled gaps, and summarised key stats?	1-15	“Do I see a clear process of cleaning and summary?”
NumPy / Numerical Analysis (20)	Do they show calculations or numerical results (tables, stats) with explanations?	1-20	“Are numbers analysed and explained?”
Visualisation (20)	Are there clear, labelled graphs or charts showing trends, patterns, and comparisons?	1-20	“Do graphs help tell the story?”
Database Integration (20)	Do they mention or show a database, queries, or importing/exporting data? Screenshots or schema diagrams count as evidence.	1-20	“Is there proof of database use?”
Python/Excel Analysis (15)	Are there charts, conditional formatting, or transformed tables visible?	1-15	“Did they use Excel or Python effectively?”
Report & Demo (10)	Is the report 7-9 pages, with datasets, methods, charts, conclusions, and a demo or screenshots?	1-10	“Can a general reader follow their write-up?”

Criteria	Excellent (8-10 / 13-15 / 17-20)	Good (5-7 / 8-12 / 10-16)	Poor (1-4 / 1-7 / 1-9)
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Q1: Data Preparation (15)	Clean data, clear handling, insightful stats	Minor issues, partial stats	Dirty data, missing key steps
Q2: NumPy Analysis (20)	Correct arrays, clear explanations	Some errors, limited insight	Incorrect or no analysis
Q3: Visualisation (20)	Clear, labelled plots, correct analysis	Basic plots, few insights	Poor or missing plots
Q4: Database (20)	Correct queries, safe updates, CSV export	Partially working, limited queries	No working database
Q5: Python/Excel Analysis (15)	Correct formulas, good charts	Some errors, minimal analysis	Incorrect or no results
Q6: Report & Demo (10)	Well-written report, clear demo	Average clarity	Missing or unclear

Criteria	Excellent	Good	Poor
Q1: Data Preparation (15)	13-15	8-12	1-7
Q2: NumPy Analysis (20)	17-20	10-16	1-9
Q3: Visualisation (20)	17-20	10-16	1-9
Q4: Database (20)	17-20	10-16	1-9
Q5: Python/Excel Analysis (15)	13-15	8-12	1-7
Q6: Report & Demo (10)	8-10	5-7	1-4

----- All the best -----